



DataDriven Floorplanning

ICCAD Problem C

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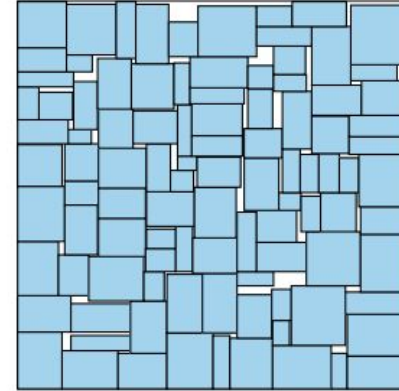
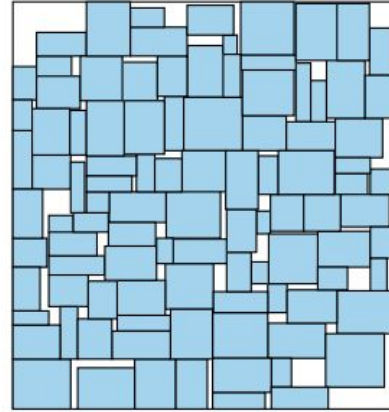
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Outline

- Problem description
- Simulated Annealing
- Previous work: GCN + RL for floorplanning
- Previous work: Deep Reinforcement Learning for Chip Floorplanning (Edge GNN + deep RL)
- CNN, GNN, and edge-GNN
- RL

Floorplanning



- Place blocks to achieve a minimum area
- Hardconstraint: Without overlap
- Soft constraint: “HPWL” & Area cost
- HPWL: Half-perimeter Wirelength

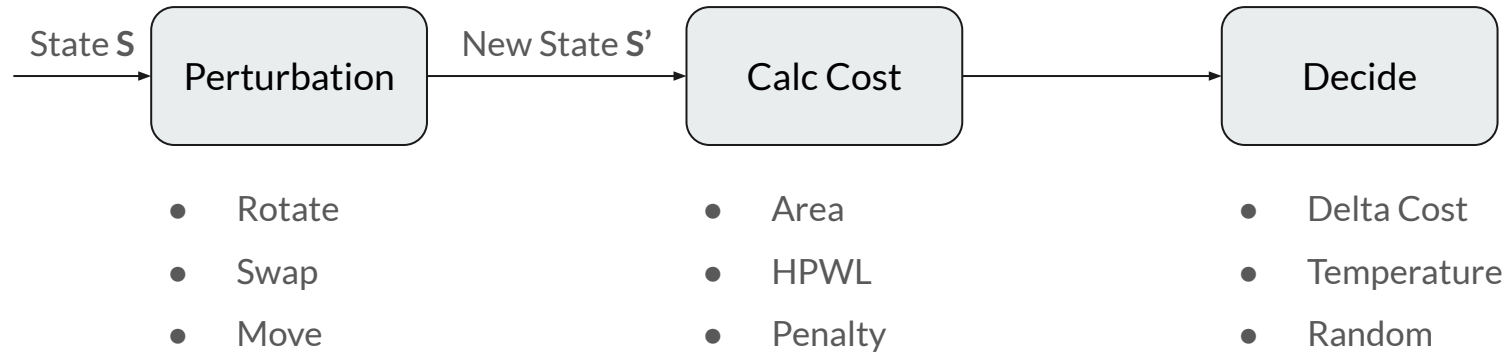
$$HPWL = (X_{max} - X_{min}) + (Y_{max} - Y_{min})$$



Problem Statement

- **Target: A feasible solution of Floorplanning with cost as low as possible**
- Hard constraints: Blocks Area & Overlap-free
- Soft constraints: Grouping, Multi-Instantiation Blocks, Boundary, Preplaced, Fixed-shape
- Cost: HPWL, Area, Runtime
- **“Suggest using ML”** (Given DataSet)

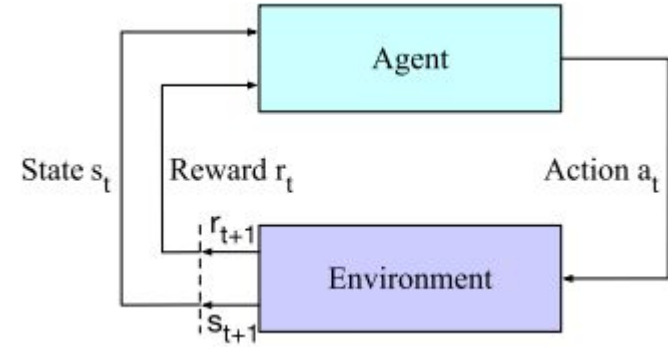
Traditional Method - Simulated Annealing



Previous Works

GCN + RL for Floorplanning [1]

- GCN(Graph Convolutional Network): Topological Structure -> Embeddings
- RL(Reinforcement Learning): Decide the next action by rewards
- Advantage: Learning instead of rolling dice





Previous Works

Deep Reinforcement Learning for Chip Floorplanning

- Problem Formulation: Markov Decision Process (MDP)
- Neural Network Architecture: Edge-GNN
- Two-Stage Placement Workflow
- Key Performance Metrics and Findings



CNN (Convolutional Neural Network)

- **Convolution Layer:** Uses kernels to capture local patterns (edges, textures) with translation invariance.
- **Pooling Layer:** Performs downsampling (Max/Avg) to reduce dimensionality and increase robustness.
- **Fully connected Layer:** Integrates all features for final classification or regression tasks.



GNN (Graph Neural Network)

- **Message Passing Layer:** Uses aggregation functions to collect neighborhood features, capturing topological relationships with permutation invariance.
- **Graph Pooling / Readout Layer:**
 - Hierarchical or global aggregation (Sum/Mean/Max)
 - Reduce graph size and generate a unified graph-level embedding.
- **Fully Connected Layer:** Integrates node-level or graph-level features for final node classification, link prediction, or regression tasks.



Edge GNN (Edge-Conditioned GNN)

- **Edge-Aware Message Passing Layer:** Incorporates multidimensional edge attributes (e.g., weights, distances) to dynamically modulate neighborhood aggregation, capturing complex interaction dynamics.
- **Edge Update Layer:** Performs feature updates on the edges themselves based on the evolving states of connected nodes to refine relational representations.
- **Fully Connected Layer:** Integrates the refined node and edge features for specialized tasks like edge classification or precise graph regression.



Reinforced Learning - Introduction

- RL trains an agent by trial and error.
- Markov Decision Process (MDP): A decision is only made base on the current state.
- Key elements of RL: States, Actions, State Transitions and Reward



Reinforced Learning - Key Elements

- **States:** The environment status, including graph embedding of the netlist, status of partial placement, etc.
- **Actions:** The decision space. Can be discrete or continuous.
- **State Transitions:** The environment's rules for updating the next state.
- **Reward:** The inverse of cost function

$$\text{Cost} = \begin{cases} \left(1 + \alpha \cdot (\text{HPWL}^{gap} + \text{Area}_{bbox}^{gap})\right) \times e^{\beta \cdot \text{Violations}^{relative}} \times \max(0.7, \text{RuntimeFactor}^{\gamma}), & \text{if feasible} \\ M, & \text{if infeasible} \end{cases}$$



Reinforced Learning

- **Zero-Shot Capability:** The ability of a pre-trained policy to instantly generate reasonable placements for previously unseen chip blocks without any fine-tuning.
- **Future Research Direction:** Optimal state representations, Discrete v.s. continuous actions



Reference

- [1] Qi Xu, Hao Geng, Song Chen, Bo Yuan, Cheng Zhuo, Yi Kang, Xiaoqing Wen (2021, Nov. 30). *GoodFloorplan: Graph Convolutional Network and Reinforcement Learning-Based Floorplanning*.
- [2] H. Ren and J. Hu, Eds., *Machine Learning Applications in Electronic Design Automation*. Cham, Switzerland: Springer Nature Switzerland AG, 2022.
- [3] A. Mirhoseini, A. Goldie, M. Yazgan, et al., "A graph placement methodology for fast chip design," *Nature*, vol. 594, no. 7863, pp. 359–364, 2021.